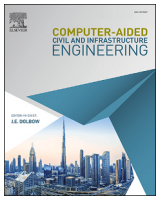
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Research Article



Early warning model for secondary collapse of semi-ruined reinforced concrete frame structures under Aftershocks

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ABSTRACT

Secondary collapse of semi-ruined reinforced concrete frame structures (RCFSSS) during aftershocks poses severe risks to rescuers. However, existing data-driven models are hindered by limited experimental data, domain discrepancies, and unstable performance under complex seismic perturbations, leaving a clear gap in achieving reliable and generalizable early-warning predictions under small-sample conditions. This study proposes a transfer-learning–guided hybrid Dilated CNN–BiLSTM–Transformer (DCBT) model that integrates extensive numerical simulations with limited high-fidelity shaking-table experiments. Predicted vertical-velocity (VV) histories are converted into cumulative absolute vertical-velocity (CAVV) curves to derive interpretable risk thresholds. Comprehensive ablation and comparative experiments show that the proposed model outperforms conventional hybrid models and baseline methods in both accuracy and robustness. Transfer learning effectively bridges simulation–experiment gaps, yielding substantial improvements in stability and generalization under varying seismic conditions. Overall, the proposed framework provides a practical and scalable early-warning tool for rapid secondary-collapse risk assessment and informed decision-making in post-earthquake rescue scenarios.

1. Introduction

Semi-ruined reinforced concrete (RC) frame structures (RCFSSS) are highly vulnerable to secondary collapse during strong aftershocks (Massumi et al., 2024), posing severe risks to rescuersl. During the 1985 Mexico City earthquake (M8.1), more than 100 rescuers and volunteers lost their lives in such collapses (Xu et al., 2023). Aftershocks occur most frequently within the critical "72-hour golden rescue" window (Baranov et al., 2023); for example, over 20 events exceeding M 5 were recorded during the 2008 Wenchuan earthquake (Guan et al., 2022). These recurrent shocks not only shorten the already limited rescue time but also increase risks to response teams, underscoring the urgent need for effective early-warning approaches specifically targeting secondary collapse of RCFSSS.

Early-warning strategies in earthquake-affected areas are generally categorized into indirect and direct approaches. Indirect approaches rely on earthquake early-warning (EEW) systems that alert rescuers before aftershocks occur. However, these systems suffer from limitations, including incomplete coverage, low sensitivity to moderate but damaging aftershocks, and intrinsic trade-offs between warning accuracy and timeliness (Saad et al., 2022; Zollo et al., 2023). Direct approaches, by contrast, monitor structural responses or shoring behavior under aftershocks—for instance, displacement and tilt measured by total stations (FEMA, 2000) or pressure monitored in timber shoring (DHS, 2011). Yet, these methods are typically localized and static, which limits their ability to provide comprehensive and real-time secondary-collapse warnings.

Given these shortcomings, conventional approaches are inadequate for predicting secondary collapse of RCFSSS under aftershocks. Machine learning has demonstrated substantial advances in risk forecasting across diverse domains—including structural systems, slopes, landslides, and heritage walls (Pratesi et al., 2015; Dick et al., 2015; Carlà et al., 2017). However, most existing models focus on gradual instabilities, whereas semi-ruined structures subjected to aftershocks may experience abrupt nonlinear failures. These failures are characterized by time-history response inflection points—including vertical displacement (VD), vertical acceleration (VA), vertical velocity (VV)—as well as rapid modal transitions (Xu et al., 2024). The urgency, sensitivity, and adaptability required for these predictions exceed the capacity of conventional ML models, underscoring the need for a specialized

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early-warning framework. Along this line, Entezami et al. (2025) proposed an early-warning strategy that couples manifold learning–aided data clustering with non-parametric probabilistic anomaly detection, illustrating how warning decisions can be made with reduced dependence on explicit collapse labels.

Nevertheless, a critical challenge lies in dataset acquisition. In real-world rescue operations, obtaining sufficient collapse data for RCFSSS to train and validate ML models is nearly impossible. Shaking table experiments (Yang et al., 2022) and numerical simulations (Sediek et al., 2021) are often used as surrogates, yet both have limitations. Shaking table tests are costly and scarce, while numerical simulations may not fully capture real-world degradation mechanisms, particularly under large deformations, strong discreteness, and complex aftershock-induced dynamic responses. Consequently, available datasets remain limited and partially inconsistent, restricting model generalization.

In summary, two critical gaps persist in assessing RCFSSS secondary collapse risk under aftershocks: (1) a mismatch between existing collapse-prediction models and the abrupt nonlinear failure characteristics of RCFSSS, and (2) the scarcity and inconsistency of datasets, which constrain predictive model generalization.

To address these gaps, this study develops a machine-learning framework tailored for secondary collapse prediction of RCFSSS under aftershocks. Long Short-Term Memory (LSTM) networks have been widely applied for dynamic response prediction in structures (Xu et al., 2020; Smrithi et al., 2022; Xiang et al., 2024). However, their reliance on fixed activation functions within Multi-Layer Perceptrons (MLPs) restricts their ability to capture highly nonlinear features (Liu et al., 2024), motivating the exploration of improved architectures (Farzad et al., 2019). Recent advances offer promising alternatives: Bidirectional LSTM (BiLSTM) networks capture temporal dependencies in both directions and have been used in structural health monitoring, such as crack propagation tracking (Wang, 2024); Convolutional Neural Networks (CNNs) extract high-frequency vibration features useful for damage detection (Tran-Ngo, 2024); and attention mechanisms dynamically weight critical features, thereby improving seismic response prediction (Lei, 2025). Building on these insights, this study introduces a hybrid Dilated CNN–BiLSTM–Transformer (DCBT) framework to effectively capture the complex and rapid collapse responses of RCFSSS under aftershocks.

Transfer learning provides a further means to mitigate data scarcity and address discrepancies between simulations and experiments. Existing applications demonstrate its versatility, including small-sample classification (Saha, 2016); structural damage identification, semi-supervised recognition, and energy prediction (Gao et al., 2018, 2021, 2022); post-earthquake damage detection using adversarial adaptation (Xu, 2021); dam structural health monitoring response prediction (Li, 2022); Bayesian updating for uncertainty reduction (Zhang et al., 2022); rapid damage classification via influence-function transfer (Lin et al., 2022); RC column capacity prediction (Pak et al., 2022); bridge response forecasting (Guo et al., 2025); PINNs-enhanced response prediction (Hu et al., 2024); nonlinear response modeling with nearest-neighbor transfer (Pak et al., 2025); and real-time ground motion inversion (Zar et al., 2025).

Inspired by recent advances, this study develops a transfer learning–guided DCBT model to address dataset scarcity and reconcile discrepancies between simulation and experimental data, thereby improving the accuracy of structural response predictions. However, directly using the predicted structural dynamic responses—such as VV curves—poses challenges for quantifying early warning indicators. To overcome this issue, the VV curves are transformed into CAVV curves, whose distinct inflection points can be identified as early warning thresholds. Based on this principle, a real-time early warning model for the secondary collapse of RCFSSS under aftershocks is established.

The major contributions of this study are threefold:

- Proposes a hybrid DCBT model to capture the urgency, sensitivity, and unpredictability of RCFSSS dynamic responses.
- Introduces a transfer-learning–enhanced framework to mitigate data scarcity and bridge gaps between simulations and shaking table tests of RCFSSS under aftershocks.
- Establishes a real-time early-warning model for RCFSSS secondary collapse based on CAVV curves, demonstrating strong practical applicability.

The paper is organized as follows. Section 2 introduces the transfer learning–guided hybrid DCBT framework. Section 3 describes the shaking table tests and numerical simulations used for dataset construction. Section 4 presents the model validation and performance analysis. Finally, Section 5 discusses risk classification and the practical applicability of scenario-based early-warning implementation.

2. Methodology

2.1. Overall framework

The real-time early-warning model for RCFSSS collapse under aftershock conditions is developed through the following steps, as illustrated in Fig. 1.

(1) High-fidelity numerical models of RCFSSS are used to generate VD, VA, and VV time-history curves for the source domain dataset, while scaled shaking table tests provide the corresponding target domain dataset.
(2) Source and target domain datasets are standardized to ensure comparability.
(3) The proposed architecture integrates Dilated CNN, BiLSTM layers, and,multi-head attention to predict VV curves of RCFSSS.
(4) The model is trained and validated. Hyperparameters are adjusted iteratively until performance criteria are satisfied.
(5) Pre-trained model weights are extracted for transfer learning.
(6) Various transfer learning strategies are employed to fine-tune the model by freezing most layers and retraining the final dense layers, together with selected DCBT layers, using target-domain data.
(7) The performance of the network is evaluated, and VV curves are predicted.
(8) Predicted VV curves are converted to CAVV curves using Eq. (1). Three risk levels—low, middle, and high—are defined, and early-warning results are automatically provided to rescue teams.

$$CAVV_i(t) = \sum_{k=1}^{i} |VV_k(t)| \tag{1}$$

where VV_k is the velocity at the k-th sampling point, the sampling interval is 0.05 s (sampling rate of 20 Hz), i is the index of the current time step ($i = 1, 2, \ldots, N$), and N is the total number of sampling points.

2.2. DCBT architecture

This study proposes a hybrid DCBT architecture (Fig. 2), implemented through a three-stage data processing pipeline. First, the dilated convolution module expands the receptive field without increasing parameters or reducing sequence length (Eq. (3)), enabling effective multi-scale extraction of temporal patterns from non-stationary sequences such as VD, VA, and VV curves. In addition, structural characteristics of the RCFSSS are incorporated, including *ECD*, defined as the ratio of the residual to the original concrete cross-sectional area of the RC column, and θ, denoting the inclination angle of the RCFSSS. Next, the BiLSTM layers capture bidirectional temporal dependencies—both short- and

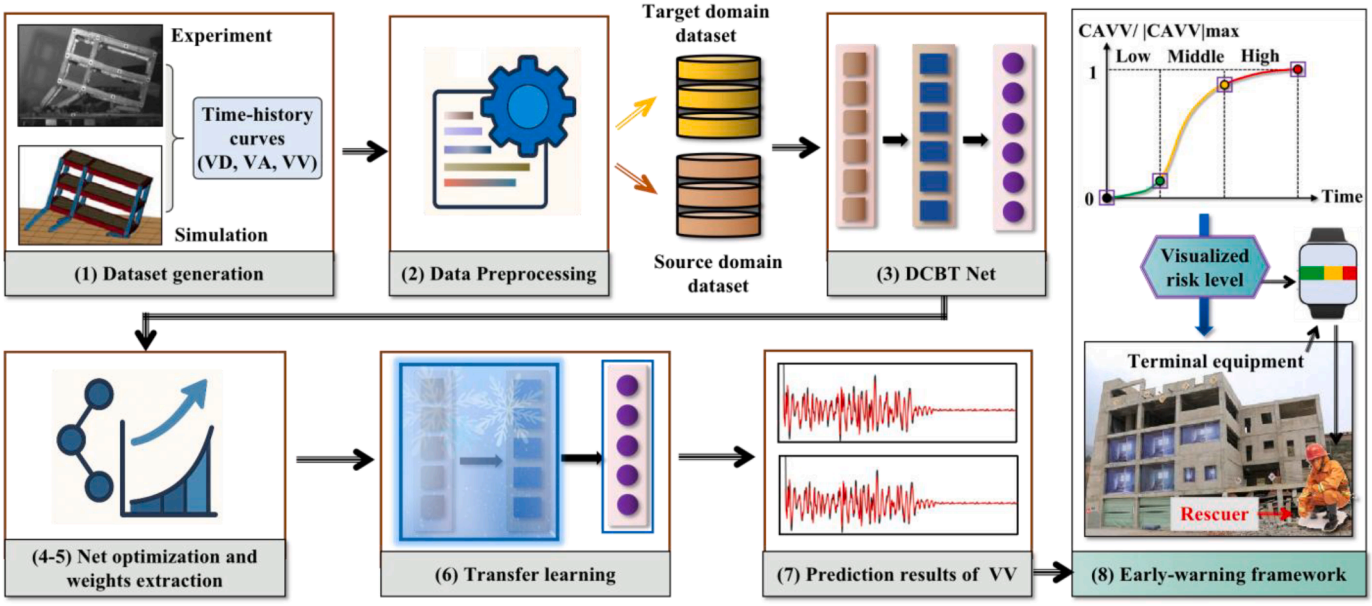


Fig. 1. Overall framework.

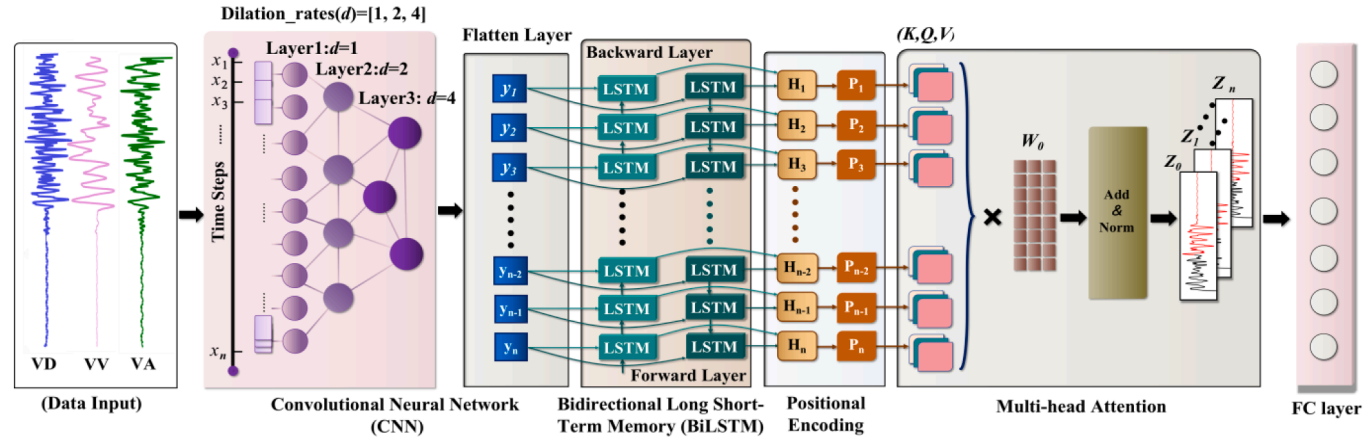


Fig. 2. DCBT model Architecture.

long-term—while mitigating gradient degradation through gated recurrent mechanisms. Finally, the extracted temporal features are passed to the Transformer module, where multi-head attention learns global relationships and long-range dependencies. The output layer then applies a linear projection to generate high-quality predictions.

(1) Dilated CNN Local Feature Extraction Module

To capture both local features and long-range dependencies in time series, the bottom-layer feature extraction network employs dilated convolution. Compared with conventional 1D convolution (Ye, 2022), dilated convolution exponentially enlarges the receptive field without increasing parameters or computational cost, enabling efficient modeling of temporal patterns across multiple time lags.

For a one-dimensional sequence $x_{(1,2,\ldots,t)}$, a dilated convolution with kernel size k and dilation rate d is defined as:

$$y(t) = \sum_{i=0}^{k-1} W_i \cdot x_t + i \cdot d \tag{2}$$

where, y (t) is the convolution output at time step t; W_i denotes the i th weight of the kernel; k is the kernel size; d is the dilation rate (reducing to standard convolution when $d = 1$); $x_{t+i \cdot d}$ is the input value sampled at intervals of d.

The effective receptive field (ERF) of the dilated convolution can be expressed as:

$$ERF = (k-1) \cdot d + 1 \tag{3}$$

where k is the temporal length (number of taps) of the 1D convolution kernel, i.e., the number of input samples used per convolution.

(2) Bidirectional LSTM Time-Series Modeling Module

CNN features are flattened into a temporal sequence y and processed by a BiLSTM. Each directional LSTM contains the standard input, forget, candidate, cell-update, and output gates. Forward and backward hidden states are concatenated to form the contextual sequence H_i. Positional encodings P_i are added to H_i to provide explicit temporal information for the subsequent attention module.

(3) Position-Aware Multi-Head Attention Module

To enhance sensitivity to temporal positions, sinusoidal positional encoding is applied:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i}/d_{\text{mod el}}}\right) \tag{4}$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i}/d_{\mathrm{mod}\ el}}\right) \tag{5}$$

A multi-head attention mechanism is then used to capture global temporal dependencies in parallel subspaces. For query Q, key K, and value V, the scaled dot-product attention for a single head is defined as

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

The multi-head attention mechanism computes attention across h heads in parallel:

$$MultiHead(Q,K,V) = Concat(head_1,\ldots,head_h)W^o \tag{7}$$

where each head is computed as:

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{8}$$

Here, W_i^Q, W_i^Q, W_i^V, and $W^O$are learnable projectionmatrices; d_k is the key dimension; and contat(·) denotes vector concatenation.

2.3. Transfer learning strategies

To overcome data scarcity and domain discrepancies between numerical simulations and shaking table tests, it is essential to employ transfer learning to improve model robustness and generalization. Accordingly, five complementary transfer learning strategies (Fig. 3 and Supplementary Figure A.1) were systematically designed and compared, each reflecting a different trade-off between computational efficiency, adaptability, and risk of overfitting:

- Strategy I – Feature Extraction: The model is pre-trained on the source domain, after which most layers are frozen and only the output layer is fine-tuned on the target domain, using the network primarily as a feature extractor with minimal computation.
- Strategy II – Full Fine-Tuning: All parameters of the pre-trained model are fine-tuned on the target domain. This allows complete adaptation but requires more computational resources and carries a higher risk of overfitting.
- Strategy III – Partial Fine-Tuning: A subset of layers (e.g., CNN frozen, BiLSTM/Transformer and output layer updated) is fine-tuned while others remain fixed. This balances computational efficiency and adaptation capability.
- Strategy IV – Training from Scratch: The model is trained solely on the target domain data without pre-trained weights. This serves as a baseline to evaluate the benefits of transfer learning.
- Strategy V – Layer-wise Fine-Tuning: Parameters are gradually unfrozen during training, typically from top layers to lower ones. This mitigates catastrophic forgetting and enhances stability.

2.4. Model evaluation

To comprehensively evaluate the predictive accuracy, generalization, and robustness of the DCBT model, a set of widely recognized statistical metrics were employed, including Mean Squared Error (*MSE*), Root Mean Squared Error (*RMSE*), Mean Absolute Error (*MAE*), Coefficient of Determination (R^2), Mean Absolute Percentage Error (*MAPE*, %), Symmetric Mean Absolute Percentage Error (*SMAPE*, %), Root Mean Squared Percentage Error (*RMSPE*, %), and Pearson Correlation Coefficient (*PCC*).

3. Experimental preparation

3.1. Data collection

3.1.1. Target domain

Two shaking-table test models were constructed to represent typical collapse modes of RC frame structures. The prototype was derived from observations of the 2008 Wenchuan M8.0 earthquake in China. The test models used M5 micro-concrete and galvanized iron wire to simulate concrete and reinforcing steel, respectively, and were tested on a small servo triaxial shaking table. The Wolong ground motion record from the 2008 Wenchuan earthquake was applied to reproduce the transition of the RC frame from an intact state to an RCFSSS.

Twenty-two representative aftershock records were selected following FEMA-P695, Appendix A (FEMA, 2009). Structural responses were captured using an AcutEye high-speed photography system, and monitoring point arrangements on the RC frame models are shown in Figs. 4(a)–(b). At each monitoring point, VD, VV, and VA time-history curves were recorded. Four sets of shaking-table tests were conducted (two representative test models are shown in Supplementary Figure A.2), each with 12 measurement points, and on average, five secondary collapses occurred per model during the 22 aftershocks. This procedure produced a total of 720 dynamic response time-history curves ($4 \times 12 \times 3 \times 5 = 720$), which constitute the target-domain dataset.

3.1.2. Source domain

A high-fidelity numerical modeling framework was developed to simulate RC frame structures with improved accuracy in capturing collapse mechanisms. Concrete and rebar were discretized separately to explicitly represent bond–slip effects at the concrete–rebar interface of RC columns, a critical factor affecting collapse prediction accuracy (Zheng et al., 2023). To balance computational efficiency with structural fidelity, co-nodes were introduced at beam–column–slab joints and along beams and slabs, thereby reproducing the monolithic behavior of cast-in-place construction (China Architecture, 2010). Concrete components were modeled using 8-node reduced-integration solid elements, whereas rebar was represented by 2-node Hughes–Liu beam elements with 2×2 Gauss quadrature integration to accurately capture bending–shear interactions. The contact_automatic_surface_to_surface contact algorithm was applied across the entire model to ensure robust interaction representation.

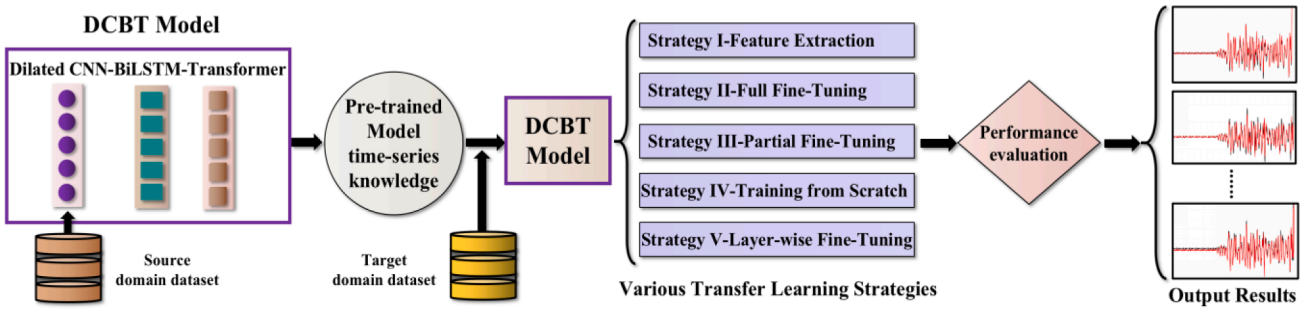


Fig. 3. Transfer learning strategies.

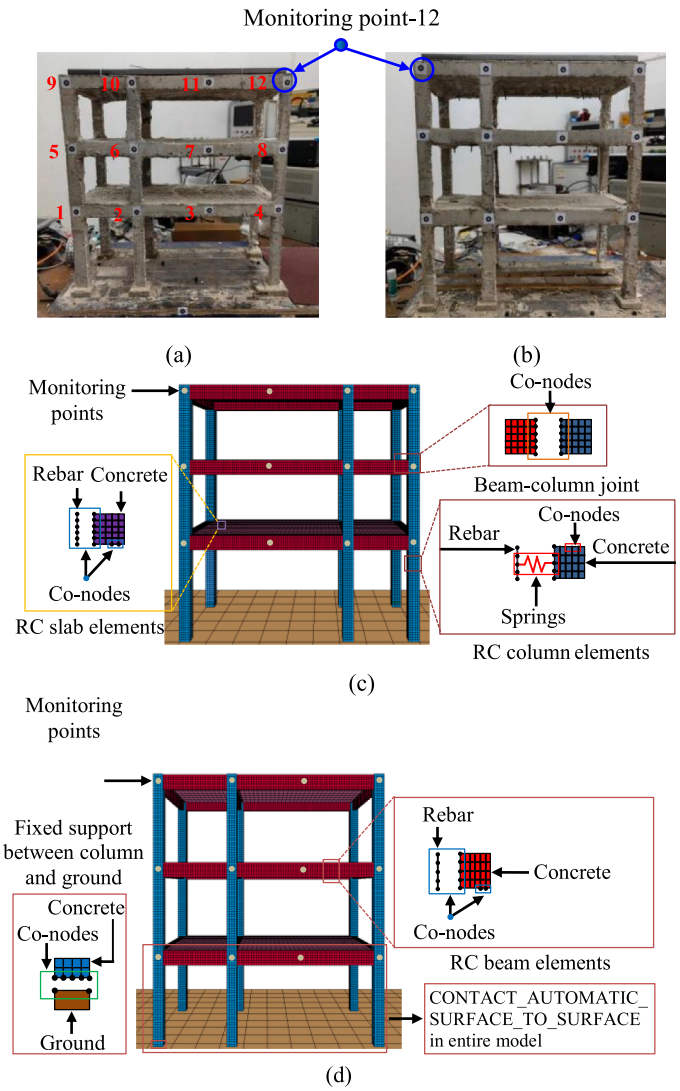


Fig. 4. RC frame structure models: (a) -(b) Experiment model-1 and −2, and (c)-(d) Simulation model-1 and −2.

Material constitutive behavior in LS-DYNA was defined as follows: concrete adopted the Karagozian & Case model (Malvar et al., 1997) to capture nonlinear crushing, tensile cracking, and strain-rate effects; rebar followed the Cowper–Symonds model (Nateghi et al., 2020) with isotropic hardening and dynamic yield scaling; and bond–slip effects were represented by spring elements at concrete–rebar nodes via the *CONTACT_1D keyword. Element deletion was applied to simulate spalling in beam, slab, and column (Xu et al., 2022). Mechanical properties of concrete and steel matched those of experimental specimens (Supplementary Tables A.1–A.3).

Validation was performed by comparing shaking-table Models-1 and −2 with their numerical counterparts under identical aftershock-induced damage states. Both experimental and numerical models exhibited three collapse phases: (I) severe plastic hinge formation at first-story columns, with story drift ratios of 0.045–0.06 (Tsinghua University, 2021), defining the RCFSSS state; (II) initial failure via long-span edge column collapse and residual deformation in short-span columns; and (III) final collapse triggered by short-span column failure. Figs. 5(a)–(F) document these progressive modes, showing close agreement between test and simulation results, thereby confirming the validity of the modeling approach using collapse mode as the evaluation index.

To further investigate the structural dynamic response of the numerical model, the time-history curves of VD, VV, and VA at monitoring point 12 of the RCFSSS-1 prototype were systematically compared with those obtained from the corresponding shaking table test during collapse progression under identical aftershocks. As shown in Figs. 6(a)–(c), the comparison reveals noticeable discrepancies in amplitude and phase between the experiment and simulation results, although the overall temporal trends exhibit considerable similarity.

To develop a collapse early-warning model for RCFSSS that is applicable to real-world rescue scenarios, expanding the structural parameter space is required to generate a comprehensive source-domain dataset. The key geometric and structural parameters are systematically presented in Table 1 and Fig. 7. When an RC frame structure reaches Phase I under mainshock, 100 typical aftershocks (Supplementary Figure A.3) are selected. By combining 64 numerical models × 100 aftershocks × 10 beam–column joint monitoring points × 3 structural dynamic responses (VA, VV, and VD), a total of 192,000 curves of the RCFSSS are obtained.

3.2. Data preprocessing

First, all records are ordered chronologically and split at the record level into 80 %/10 %/10 % training, validation and test sets. A sliding-window scheme is then applied only within each subset: the raw time histories are segmented with a window length of $W = 120$ steps (≈1.2 s) and a stride of $S = 10$ steps (91.7 % overlap). Each channel is normalized

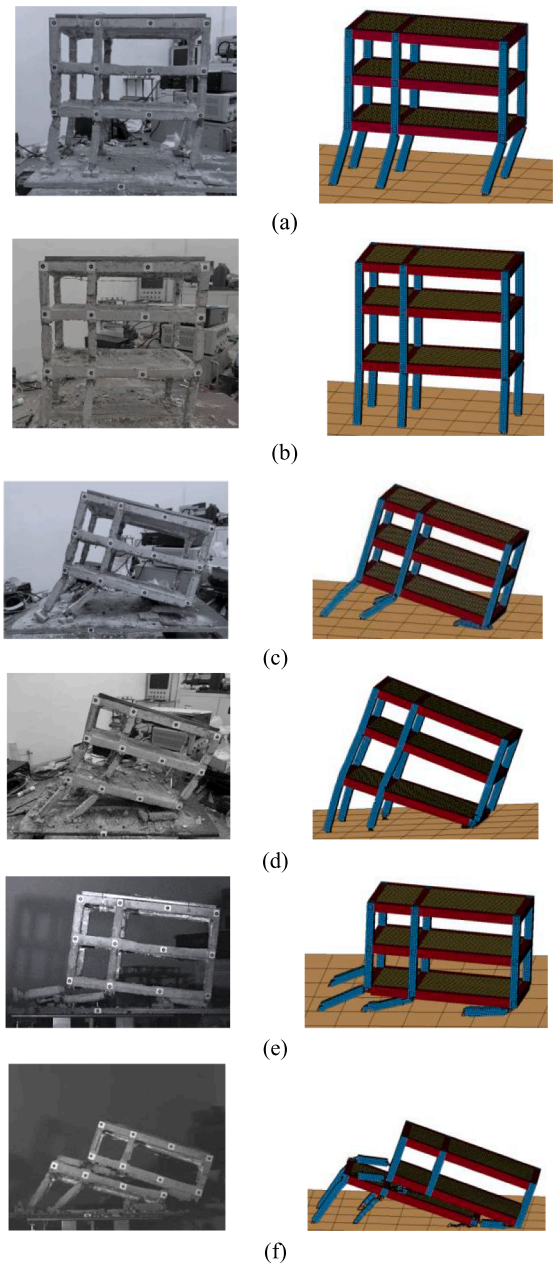


Fig. 5. Comparison of collapse mode between experiment and simulation: (a)-(b) pre-collapse, (c)-(d) RCFSSS-1and −2, and (e)-(f) secondary collapse of RCFSSS-1and −2.

according to Eq. (9).

$$X_{norm(i)} = \frac{X_{(i)} - \mu_{(i)}}{\sigma_{(i)}} \tag{9}$$

where, $\mu_{(i)}$ and $\sigma_{(i)}$ denote the mean and standard deviation of channel $i \in \{VD, VV, VA\}$.

3.3. Computational environment

Training used 2 × NVIDIA A100 GPUs, an AMD EPYC 7742 CPU, 512 GB RAM, and NVMe SSD RAID 0, with PyTorch 1.12.1, CUDA 11.6, mixed-precision, NCCL 2.16.2, and TensorBoard 2.10.0 for visualization.

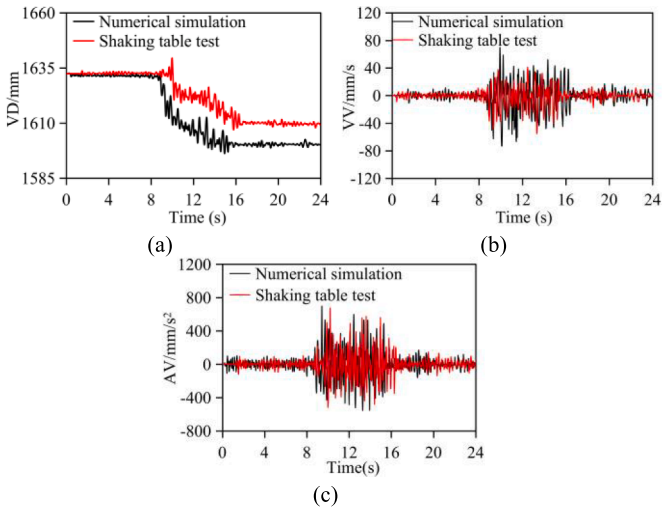


Fig. 6. Comparison of structural dynamic response: (a) VD curves, (b) VV curves, and (c) VA curves.

4. Experimental results

4.1. The combination of DCBT model

To determine the optimal architecture, various layer-sequencing strategies of the proposed DCBT model were systematically evaluated with attention head settings of 4, 8, and 16. Other hyperparameters were automatically optimized using Optuna and Ray including dropout, dilation blocks, learning rate, batch size, epochs, optimize (see Supplementary Table A.4). Numerical simulation data were split into an 80 % training set and a 20 % test/validation set. Model performance was assessed using the eight evaluation metrics summarized in Section 2.4, as well as computational time (CT). All experiments were conducted on the numerical domain, and the reported results correspond to the testing phase (Table 2).

Table 2 summarizes the comparative results for different layer sequencing and configurations of the DCBT model, from which the following findings can be drawn:

- The hybrid DCBT sequential architecture consistently outperformed other configurations. In particular, the 2 BiLSTM + Transformer variant (Heads=4) achieved the lowest *RMSE* (112.66) and *MAE* (36.84), alongside the highest R^2 (0.871) and *PCC* (0.93).
- Adding more than two BiLSTM layers did not enhance predictive accuracy. For instance, moving from 2 to 3 BiLSTM layers resulted in reduced accuracy (R^2 dropping to 0.759–0.819) and increased computational time (3.6–4.8 h). With 4 BiLSTM layers, performance deteriorated further (R^2 = 0.624–0.669) despite even longer training times.
- The number of attention heads also influenced outcomes. A moderate head count (4) provided the best balance between accuracy and computational cost. Larger head counts (8 or 16) increased training time (by up to ~40 %) without consistent accuracy gains; in some cases (e.g., 2 BiLSTM, 16 heads), performance even declined.
- The 2 BiLSTM + 4-head Transformer configuration offers the optimal trade-off between prediction accuracy and computational efficiency for modeling the dynamic responses of RCFSSS under aftershocks.

4.2. Hyperparameter settings

The hybrid DCBT model is highly sensitive to hyperparameters such as kernel size, dilation rate, learning rate, and dropout ratio. To reduce computational cost, the analysis focused on the optimized configuration (2 BiLSTM layers, 4 attention heads; Table 2). The model was retrained

Table 1
Geometric and structural information for RC frame structures.

L_1/mm	L_2/mm	L_3/mm	H_1/mm	H_2/mm	n	A/mm	B/mm	C/mm	D-1/mm	D-2/mm
3000	4500	6000	4000	3300	2	500	250	550	20	18
	6000	7200		3600	5	550	300	650	22	20
	8100				8	600	350	750	24	22

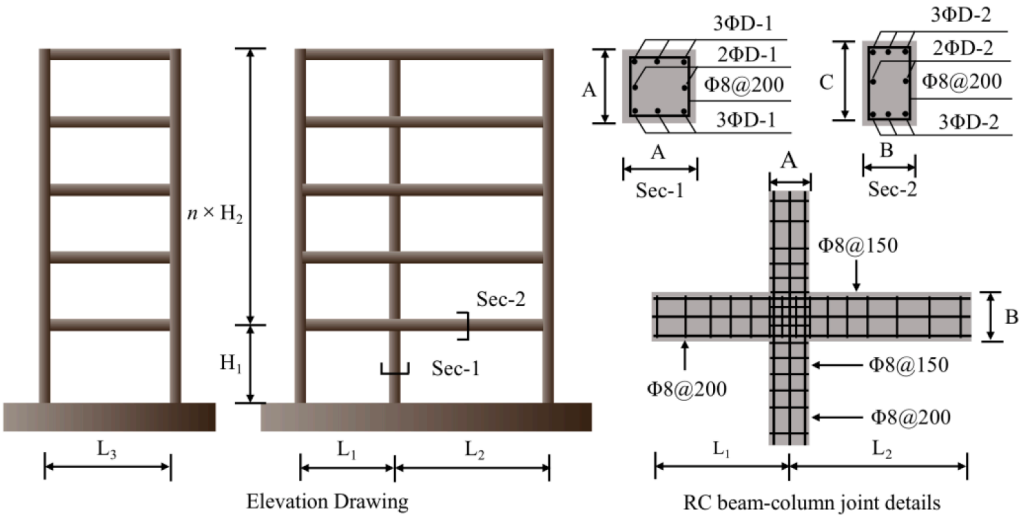


Fig. 7. Geometric information for RC frame strcture.

Table 2
Comparative performance of DCBT model configurations.

BiLSTM #	Heads	*MSE*	*RMSE*	*MAE*	R^2	*MAPE*	*SMAPE*	*RMSPE*	*PCC*	*CT(h)*
1 ×	4	12,879	113.49	38.25	0.849	34.91	24.31	32.65	0.92	2.4
	8	19,180	138.52	40.42	0.771	36.63	28.07	31.67	0.79	3.8
	16	17,198	131.14	42.70	0.798	38.84	25.98	32.30	0.90	4.4
2 ×	4	12,692	112.66	36.84	0.871	44.18	27.26	30.84	0.93	3.3
	8	17,384	131.85	43.18	0.796	42.44	25.95	39.61	0.89	4.0
	16	13,295	115.31	39.02	0.844	35.48	22.27	26.11	0.92	4.7
3 ×	4	20,578	143.45	47.72	0.759	40.17	26.79	29.73	0.87	3.6
	8	24,315	155.93	52.72	0.715	38.97	25.64	28.21	0.85	4.4
	16	15,411	124.14	41.24	0.819	37.14	26.69	22.79	0.91	4.8
4 ×	4	31,085	176.33	64.53	0.624	42.81	26.92	23.64	0.69	3.9
	8	29,917	172.97	62.08	0.649	40.52	30.88	21.38	0.81	4.5
	16	28,255	168.09	60.55	0.669	42.24	32.18	24.72	0.83	4.9

with varying kernel size, dilation rate, dropout, and learning rate, while other parameters were auto-tuned via Optuna and Ray(see Supplementary Table A.4). The results are shown in Supplementary Figure A.4, highlighting the hyperparameter sensitivity of the DCBT model.

- Kernel size: As a key CNN parameter, larger sizes generally improved R^2, with performance peaking at size 7.
- Dilation rate: The dilation rate also plays an important role in CNN feature extraction. While accuracy (R^2) remained relatively stable as the input window size increased, it improved with larger dilation rates. The configuration dilation rate = [1, 2, 4, 8] achieved both high and stable performance.
- Dropout ratio: With dropout values between 0 and 0.5, model accuracy (R^2) remained stable. Beyond 0.5, performance declined noticeably. The best accuracy was observed with dropout < 0.5 combined with 4 attention heads
- Learning rate: Model accuracy (R^2) increased as the batch size grew from 1 to about 180, but declined beyond this point. Meanwhile, higher learning rates consistently reduced accuracy. The optimal setting was found at learning rate = 0.0001.

4.3. Ablation studies

Based on the optimal DCBT parameters obtained in Sections 4.1 and 4.2, ablation studies were conducted to further validate the model architecture. Three simplified variants—BiLSTM–Transformer, CNN–BiLSTM, and CNN–Transformer, CNN-LSTM, LSTM-Transformer, 1D-CNN-GRU-LSTM, and Transformer-GRU-BiLSTM—were compared with the full DCBT model in predicting VV curves at measurement point 12 during the secondary collapse of RCFSSS under aftershocks. The evaluation metrics are summarized in Table 3, and the corresponding prediction curves are shown in Fig. 8.

From Table 3, the comparative performance of various hybrid models can be summarized as follows:

- DCBT delivers the overall best performance, with the lowest errors (*MSE* = 12,532, *RMSE* = 111.81, *MAE* = 36.21) and the highest correlation metrics (R^2 = 0.875, *PCC* = 0.94). This confirms its strong capability to simultaneously capture local spatial features, temporal dependencies, and global correlations of VV responses.

Table 3
The evaluation indices of various hybrid models (Unit: mm).

Model Types	*MSE*	*RMSE*	*MAE*	R^2	*MAPE*	*SMAPE*	*RMSPE*	*PCC*
DCBT	12,532	111.81	36.21	0.875	8.13	6.95	10.45	0.93
BiLSTM-Transformer	16,877	129.60	40.96	0.812	12.44	12.61	15.93	0.87
CNN-BiLSTM	20,567	143.23	44.12	0.778	15.46	13.95	17.85	0.82
CNN-Transformer	14,810	121.65	38.54	0.826	9.94	9.64	12.74	0.91
CNN-LSTM	15,806	125.74	40.03	0.829	10.74	10.25	13.53	0.89
LSTM-Transformer	17,353	131.52	41.14	0.818	12.12	11.62	14.62	0.88
1D-CNN-GRU-LSTM	15,307	123.73	38.97	0.833	8.94	8.35	11.74	0.90
Transformer-GRU-BiLSTM	15,887	125.71	40.03	0.829	10.76	10.23	13.57	0.89

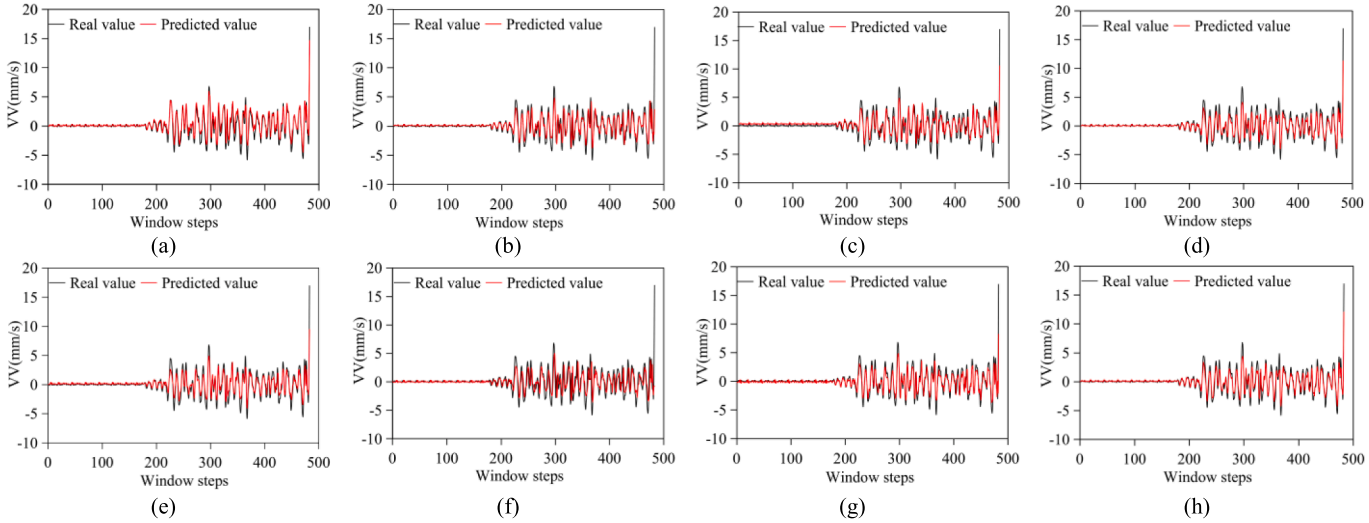


Fig. 8. Visualization of VV results: (a) DCBT, (b) BiLSTM–Transformer, (c) CNN–BiLSTM, (d) CNN–Transformer models, (e) CNN-LSTM, (f) LSTM-Transformer, (g) 1D-CNN-GRU-LSTM, and (h) Transformer-GRU-BiLSTM.

- BiLSTM–Transformer performs moderately well ($R^2 = 0.812$, $PCC = 0.87$). Although the temporal–attention hybrid enhances feature interaction, the absence of CNN-based local extraction weakens its ability to handle high-frequency fluctuations, leading to larger errors ($RMSE = 129.60$, $MAE = 40.96$).
- CNN–BiLSTM exhibits the weakest correlation ($R^2 = 0.778$) and highest *RMSE* (143.23). Without a global attention mechanism, it effectively captures local and sequential features but fails to represent long-range temporal dependencies, resulting in less stable VV predictions.
- CNN–Transformer achieves relatively balanced performance ($R^2 = 0.826$, $RMSE = 121.65$, $MAE = 38.54$, $PCC = 0.91$). The combination of CNN-based local extraction and Transformer-based global attention provides strong generalization, even though the lack of recurrent sequence modeling slightly limits its fine-grained temporal adaptation.
- Other composite models (e.g., Transformer-GRU-BiLSTM, LSTM–Transformer, 1D-CNN-GRU-LSTM,) demonstrate intermediate accuracy ($R^2 \approx 0.818$–0.833), suggesting that while hybridization improves robustness, redundant sequential modules or insufficient feature fusion may dilute learning efficiency.

4.4. Comparison with baselines

To further evaluate the performance, the DCBT model was compared with commonly used time-series forecasting models, including RNN, GRU, TCN, Informer, and Autoformer (Vardaan, K., 2019; Rofii et al., 2024; Hou, 2023; Zhou, 2021; Wu, 2021). The optimal hyperparameters for the RNN, GRU, and TCN models were selected automatically. Evaluation metrics are summarized in Table 4, and the corresponding prediction curves are visualized in Fig. 9.

From Table 4, several conclusions can be drawn regarding the comparative performance of the evaluated models:

- DCBT achieves the best overall performance, yielding the lowest *MSE*, *RMSE*, and *MAE*, along with the highest R^2 and *PCC* values.

Table 4
The evaluation indices of various baselines (Unit: mm).

Model Types	*MSE*	*RMSE*	*MAE*	R^2	*MAPE*	*SMAPE*	*RMSPE*	*PCC*
DCBT	12,532	111.8	36.2	0.875	8.13	6.95	10.4	0.93
BiLSTM	16,188	126.9	40.9	0.809	6.95	6.52	9.81	0.89
LSTM	18,600	136.5	47.5	0.761	6.96	6.34	9.05	0.87
Transformer	17,354	130.1	44.5	0.791	6.25	6.43	9.73	0.88
RNN	20,630	143.5	53.5	0.747	9.07	6.14	9.62	0.87
GRU	22,456	149.6	47.2	0.737	5.73	6.03	8.71	0.86
TCN	25,696	159.9	48.3	0.705	5.36	5.78	7.13	0.84
Informer	15,850	125.5	41.4	0.812	6.02	6.31	9.52	0.9
Autoformer	15,003	122.5	40.0	0.826	5.91	6.22	9.41	0.91

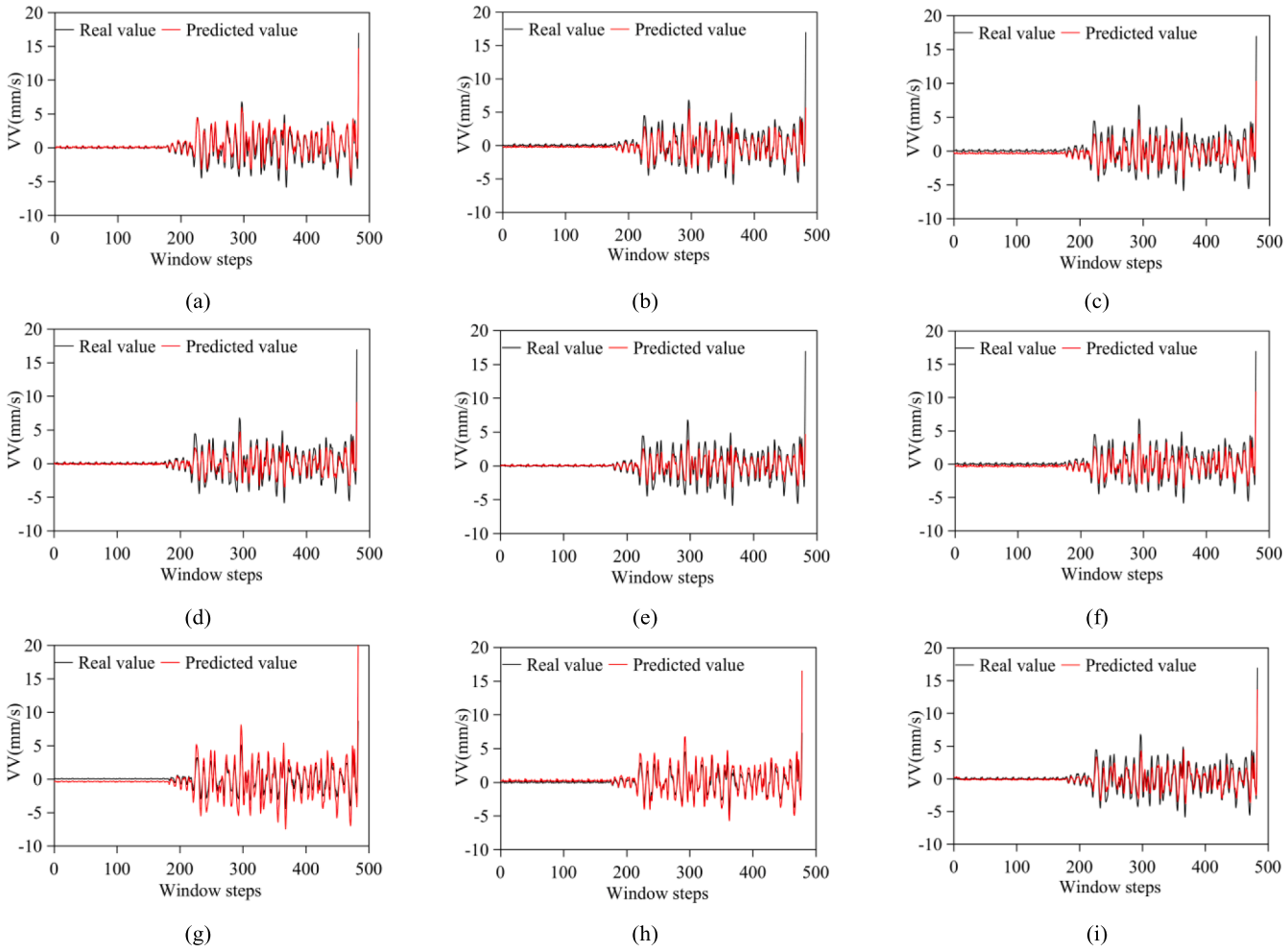


Fig. 9. Comparison of real and predicted values: (a) DCBT, (b) BiLSTM, (c) LSTM, (d) Transformer, (e) RNN, (f) GRU, (g) TCN, (h) Informer, and (i) Autoformer model.

This demonstrates its strong capability to jointly capture local, sequential, and global dependencies in complex structural dynamic responses.

- Informer and Autoformer show improved accuracy compared with traditional recurrent models, benefiting from efficient attention mechanisms and temporal decomposition; however, their performance still falls short of DCBT due to weaker local feature extraction.
- BiLSTM and Transformer models achieve moderate accuracy, indicating that single-mechanism architectures—either sequential or attention-based—are insufficient for fully representing nonlinear secondary collapse dynamics.
- RNN, GRU, and TCN models exhibit higher errors and lower R^2, confirming their limited ability to model multi-scale temporal features and abrupt transients inherent in aftershock responses.

Overall, the DCBT architecture, through its domain-specific integration of dilated convolution, bidirectional temporal modeling, and transformer-based global attention, demonstrates the most robust and accurate prediction of VV time-history curves among all baselines.

4.5. Generality evaluation

To assess the generalization and robustness of the DCBT model, a blind test was conducted on an RCFSSS configuration excluded from training. Its geometric and structural parameters are in Table 5, and the semi-ruined and secondary-collapse states in Fig. 10. Table 6 shows the

Table 5
Geometric parameters for blind-test model (Unit: mm).

L_1/mm	L_2/mm	L_3/mm	H_1/mm	H_2/mm	N	θ/°	ECD/ %
4500	3000	7200	3300	3300	3	36	68

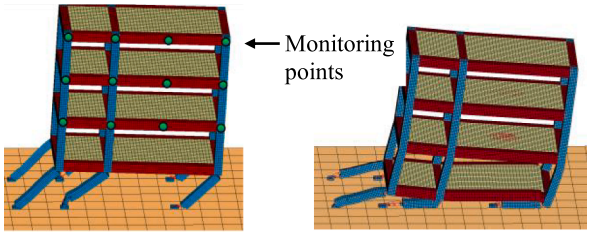


Fig. 10. Secondary collapse process of RCFSSS: (a) semi-ruined state and (b) secondary collapse state.

Table 6
The evaluation indices of BFMS-SR model.

MSE	*RMSE*	*MAE*	R^2	*PCC*	*MAPE*	*SMAPE*
12,511	113.75	36.53	0.882	0.94	8.22	7.53

results: $R^2 = 0.880$, $PCC = 0.943$, and $RMSE = 42.77$ mm, with a blind-test accuracy of 88 %. These findings confirm that DCBT reliably predicts VV curves for unseen structures, demonstrating strong generalization for post-earthquake secondary-collapse early-warning.

4.6. Transfer-learning strategies

According to the five transfer-learning strategies described in Section 2.3, the Hybrid DCBT network is used for domain adaptation and evaluated with eight performance metrics. To mimic real-world uncertainties—including sensor jitter, dropouts, lighting variability, and occlusions—band-limited Gaussian noise at 5 %, 10 %, and 15 % is added to the VD, VV, and VA curves. Each experiment is repeated 20 times with different random seeds, and the mean ± standard deviation of all metrics is reported. The results are summarized in Table 7.

Table 7 summarizes the performance of five transfer-learning strategies under perturbation levels of 5 %, 10 %, and 15 %. As perturbation increases, error metrics—including *MSE, RMSE, MAE, MAPE, SMAPE*, and *RMSPE*—tend to rise, while correlation metrics (R^2 and *PCC*) gradually decrease, indicating performance deterioration under stronger data interference. At 5 % perturbation, Strategy-1 and Strategy-3 exhibit the lowest errors (*MSE*: 2437/2439; *RMSE*: 49.12/49.49; *MAE*: 19.50/19.62) and the highest correlation (R^2: 0.952/0.951; PCC: 0.982/0.981), whereas Strategy-2 and Strategy-5 show comparatively higher errors. At 10 %, Strategy-1 maintains stable performance (*MSE*: 2638; *RMSE*: 51.35; *MAE*: 20.12; R^2: 0.948; *PCC*: 0.974), while Strategies 2–4 experience more pronounced error growth and slight reductions in correlation. Under 15 % perturbation, all strategies experience noticeable degradation, with Strategy-4 and Strategy-5 exhibiting the largest error increases and the lowest correlation values (*MSE*: 3455/3420; *RMSE*: 58.85/56.85; *MAE*: 24.35/25.35; R^2: 0.924/0.933; *PCC*: 0.961/0.968), while Strategy-1 continues to show the highest robustness.

5. Discussion

5.1. Performance advantages of the DCBT model

The DCBT model consistently outperforms other hybrid and baseline architectures in predicting the dynamic responses of RCFSSS under abrupt and highly unpredictable aftershock excitations. As shown in Tables 3 and 4, DCBT achieves the lowest error metrics (*RMSE* ≈ 111.8–112.7 mm) and the highest correlation scores ($R^2 \approx 0.875$, $PCC \approx 0.93$), demonstrating superior fidelity in reproducing secondary-collapse trajectories.

These advantages originate from the complementary strengths of its three major components: (1) Dilated CNN, which extracts multi-scale, high-frequency, and damage sensitive local patterns, enabling accurate detection of abrupt changes and inflection points in VV time-history data (CVAA) with relatively few parameters (Kiranyaz et al., 2021). (2) BiLSTM, which provides bidirectional temporal modeling, strengthening contextual representation around rapid transitions and preserving long-term phase information essential for collapse prediction under sequential shocks. (3) Multi-head self-attention (Transformer), which adaptively re-weights temporal features to capture global correlations and nonlinear, long-range interactions, thereby improving robustness while reducing dependence on large labeled datasets.

Together, these modules enable DCBT to capture the urgency (short, strong transients), sensitivity (small but informative high-frequency variations), and unpredictability (nonlinear and nonstationary shifts) that characterize RCFSSS responses under aftershocks—capabilities that single-paradigm models typically lack. Hybrid models without CNN components (e.g., BiLSTM–Transformer) fail to identify high-frequency structural response(e.g., VV time-history data) features, while architectures lacking global attention (e.g., CNN–BiLSTM) struggle to represent long-range dependencies, resulting in higher errors and reduced R^2. By integrating convolutional, recurrent, and attention mechanisms within a compact 2-BiLSTM + 4-head Transformer configuration, the DCBT model achieves an effective balance between accuracy and computational efficiency, making it well-suited for near-real-time secondary-collapse early warning in post-earthquake ruin environments Fig. 11 and 12.

5.2. Transfer learning in early-warning

A major challenge in deploying the proposed framework is the scarcity of shaking-table data for RCFSSS subjected to aftershock-induced secondary collapse. These tests are limited in scale, and real post-earthquake rescue environments introduce additional field variability—including lighting changes, occlusion, sensor dropouts, and mounting jitter—that further degrade measurement quality. Thus, shaking-table records serve as the target domain, while large-scale

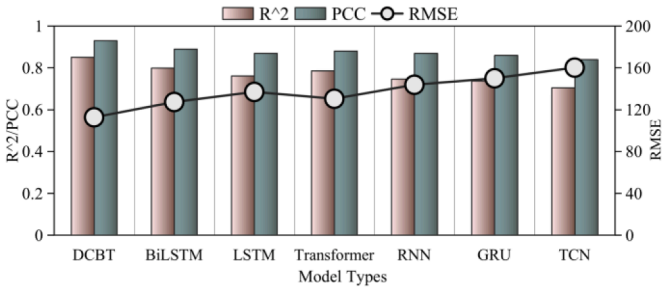


Fig. 11. Various model performance in predicting VV curves of RCFSSS under aftershocks.

Table 7
Evaluation indices (Mean ± Standard Deviation) under varying perturbation levels (mm).

Perturbation Level (%)	Types	*MSE*	*RMSE*	*MAE*	R^2	*PCC*	*MAPE*	*SMAPE*	*RMSPE*
5	Strategy-1	2433 ± 122	49.89 ± 2.63	19.51 ± 1.83	0.958 ± 0.015	0.984 ± 0.013	9.56 ± 1.16	6.43 ± 1.24	10.43 ± 2.31
	Strategy-2	2663 ± 115	51.24 ± 2.68	20.33 ± 1.85	0.942 ± 0.013	0.976 ± 0.012	10.35±1.32	6.22± 1.32	11.52 ± 2.44
	Strategy-3	2411 ± 114	49.33 ± 2.61	19.62 ± 1.87	0.956 ± 0.014	0.982 ± 0.012	9.64±1.45	6.02 ± 1.37	11.64 ± 2.55
	Strategy-4	2793 ± 132	52.43 ± 2.73	21.15 ± 1.92	0.942 ± 0.014	0.975 ± 0.012	10.43 ± 1.42	6.47± 1.35	12.33 ± 2.78
	Strategy-5	2834 ± 136	53.75 ± 3.32	21.63 ± 2.04	0.945 ± 0.015	0.972 ± 0.011	10.53 ± 1.85	7.32 ± 1.40	11.88 ± 2.42
10	Strategy-1	2662 ± 137	51.32±3.25	20.12 ± 1.98	0.942 ± 0.016	0.976 ± 0.015	9.53 ± 1.43	7.03± 1.42	12.25 ± 3.63
	Strategy-2	2877 ± 144	52.77 ± 3.16	21.94 ± 2.12	0.942 ± 0.017	0.971 ± 0.015	8.93 ± 1.21	7.84 ± 1.50	10.99± 3.46
	Strategy-3	2722 ± 135	52.32 ± 3.01	20.85 ± 2.05	0.939 ± 0.018	0.966 ± 0.016	9.02 ± 164	7.93 ± 1.55	11.16 ± 3.32
	Strategy-4	3005 ± 142	54.53 ± 3.47	22.65 ± 2.26	0.934 ± 0.019	0.965 ± 0.017	9.65 ± 1.03	8.03 ± 1.60	11.03 ± 3.55
	Strategy-5	2744 ± 131	52.32 ± 3.19	23.12 ± 2.27	0.945 ± 0.017	0.975 ± 0.014	10.23 ± 1.41	8.11 ± 1.65	12.55 ± 3.74
15	Strategy-1	2864 ± 153	53.26 ± 3.83	20.75 ± 2.55	0.945 ± 0.018	0.972 ± 0.017	11.56 ± 1.43	8.03 ± 1.65	12.46 ± 4.74
	Strategy-2	3063 ± 133	56.55 ± 3.94	22.23 ± 2.75	0.933 ± 0.019	0.963 ± 0.018	10.34 ± 1.74	7.96 ± 1.70	12.46 ± 4.67
	Strategy-3	3203 ± 124	56.66 ± 4.02	22.62 ± 2.72	0.932 ± 0.020	0.964 ± 0.018	11.34 ± 1.05	7.98 ± 1.78	12.74 ± 4.64
	Strategy-4	3433 ± 146	58.36 ± 4.21	24.35 ± 2.85	0.925 ± 0.021	0.962 ± 0.019	12.64 ± 2.11	7.92 ± 1.85	11.35 ± 4.74
	Strategy-5	3404 ± 153	56.97 ± 4.17	25.35 ± 2.86	0.935 ± 0.020	0.966 ± 0.018	13.30 ± 1.65	7.54 ± 1.90	12.24 ± 4.65

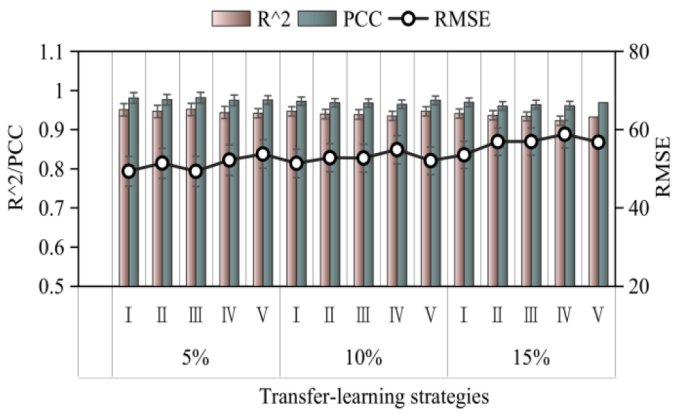


Fig. 12. Performance of five transfer-learning strategies for RCFSSS responses under aftershocks.

numerical simulations constitute the source domain. Directly training the DCBT model on small experimental datasets leads to overfitting, whereas simulation-only training suffers from domain mismatch. Transfer learning (TL) therefore provides a principled way to bridge this gap (Pan, 2010).

To evaluate TL robustness under realistic conditions, five TL strategies were tested on the DCBT architecture using laboratory data contaminated with 5–15 % field-type disturbances (Table 7). Overall, Strategy-1 (feature extraction) delivers the highest accuracy across noise levels ($R^2 \approx$ 0.95 at 5 % and $\approx$ 0.94 at 15 %). Strategy-3 (partial fine-tuning) performs similarly under low disturbance but degrades rapidly as noise increases. Strategies-2 and −5 exhibit intermediate stability, whereas Strategy-4 (training from scratch) performs worst due to data scarcity and noise sensitivity.

Under low disturbance (5 %), the domain gap is governed mainly by modeling differences, while the VV curves retain a high signal-to-noise ratio. The pretrained DCBT encoder already captures multi-scale vibration patterns, temporal dependencies, and early collapse precursors that remain valid in the target domain. Consequently, freezing most encoder layers (Strategy-1) preserves these robust features and achieves the best performance. Excessive fine-tuning (Strategies-2 and −5) risks overwriting well-learned representations, and Strategy-4 fails to learn collapse dynamics with limited data.

Under higher disturbance (10–15 %), the domain gap becomes dominated by noise-induced distortions that obscure short-lived precursor signatures. Frozen representations (Strategy-1) lose adaptability, and partial fine-tuning (Strategy-3) is insufficient to correct deeper distortions in attention-based layers. Strategy-4 continues to underperform. In contrast, Strategy-5 (layer-wise fine-tuning) becomes more competitive because its gradual unfreezing selectively adapts domain-sensitive upper layers while preserving low-level Dilated CNN filters that encode generic vibration characteristics, yielding a better balance between robustness and adaptability.

Mechanistically, TL enables the DCBT model to reuse pre-learned multi-scale vibration features, temporal dependencies, and long-range correlations from the simulation domain, allowing it to remain sensitive to the urgency (short, strong transients), sensitivity (small high-frequency changes), and unpredictability (non-stationary shifts) of RCFSSS responses—even when experimental data are scarce and degraded by field variability. Consistent with recent SHM developments (Gardner et al., 2020; Pezeshki, Hadi and Hojjat Adeli, 2023; Figueiredo et al., 2023), these findings show that TL is a core enabler for reliable early-warning systems under realistic post-earthquake conditions.

5.3. Practical early-warning framework

An early-warning model application framework for various structural types (Fig. 13(a)–(b)) was established, consisting of four synergistic phases: (1) strategic deployment of sensors at critical beam–column joints to capture essential structural dynamics (Code for Seismic Design of Buildings, 2010); (2) acquisition of high-fidelity VV data via drone-mounted high-speed cameras integrated with Kalman-ANN filtering and wireless transmission (Ribeiro et al., 2021); (3) Predictive modeling using the DCBT model with transfer-learning strategies to overcome limited experimental datasets and simulation–experiment discrepancies; and (4) rescue-centric decision support through wearable interfaces displaying real-time risk levels.

Although the proposed model can accurately predict VV curves, their direct interpretation is not intuitive for non-specialist rescuers operating under high-pressure field conditions. To address this issue, the cumulative absolute velocity (CAV) time-history curve—widely recognized for its distinct inflection points in measuring ground motion intensity (Kuran et al., 2023)—is referenced. Importantly, in bottom-frame masonry semi-ruined structures, secondary collapse is dominated by vertical deformations, making the CAVV particularly informative. The CAVV time-history curves exhibit four distinct phases: stabilization, structural response progression, secondary collapse, and rebalancing.

Based on this, the VV curves derived from various structural parameters (Table 1) under multiple aftershocks were converted into normalized CAVV curves for further analysis. A total of 64 models × 100 aftershocks × 1 beam–column joint point resulted in 6400 CAVV curves (Fig. 14). These curves, representing cumulative structural responses, exhibited two distinct inflection points that provided clearer early-warning indicators. For quantitative early-warning index analysis, statistical results showed that the first inflection point (Point A) had lower, mean, and upper values of approximately 570 mm/s, 1160 mm/s, and 1890 mm/s, respectively. Similarly, the second inflection point (Point B) had corresponding values of approximately 3950 mm/s, 5035 mm/s, and 6130 mm/s, respectively.

For practical application, the pre-secondary collapse process between the two inflection points is simplified into three stages: stabilization, structural response development, and secondary collapse (Fig. 15). The corresponding CAVV-based risk levels—O–A, A–B, and B–C—represent low, medium, and high-risk zones, respectively (Fig. 15). A continuous CAVV risk coefficient (0–1) quantifies proximity to secondary collapse, offering rescuers direct and actionable feedback. At monitoring point 12, the predicted VV and derived CAVV curves (Fig. 16) link model predictions to operational risk, illustrating how the CAVV framework supports real-time rescue decisions. Moreover, to capture the physical responses of the RCFSSS under aftershock excitations, VD and tilt angle (θ) are introduced as key indicators. The crossing of CAVV thresholds is thus correlated with variations in VD and θ, reinforcing the framework's physics-aware capability (Fig. 17).

Following the "strong joints, weak components" and "strong column/weak beam" principles (Code for Seismic Design of Buildings, 2010), upper beam–column joints in semi-ruined RC frames demonstrate greater resistance to aftershock-induced damage, as verified through field investigations, shaking table tests, and numerical simulations (Xu et al., 2022) (Supplementary Figure 5(a)–(c)). During testing, the upper joints remained largely intact, while the lower components exhibited pronounced nonlinear deformation and vertical displacement. Accordingly, monitoring points were installed at the upper joints to record VD, VV, and VA time histories, thereby minimizing data loss and improving signal fidelity. Their elevated placement also mitigated interference from collapsing lower components, further enhancing measurement reliability (Supplementary Figure 5(d)).

Collectively, this framework shows how integrating high-fidelity response prediction, CAVV-based risk quantification, and wearable interfaces bridges the gap between laboratory datasets and real-world earthquake rescue. By translating complex dynamics into intuitive risk levels, it enables near real-time, user-friendly monitoring of secondary collapse risks, supporting safer and more efficient interventions. Its applicability also extends to other semi-ruined structures, such as

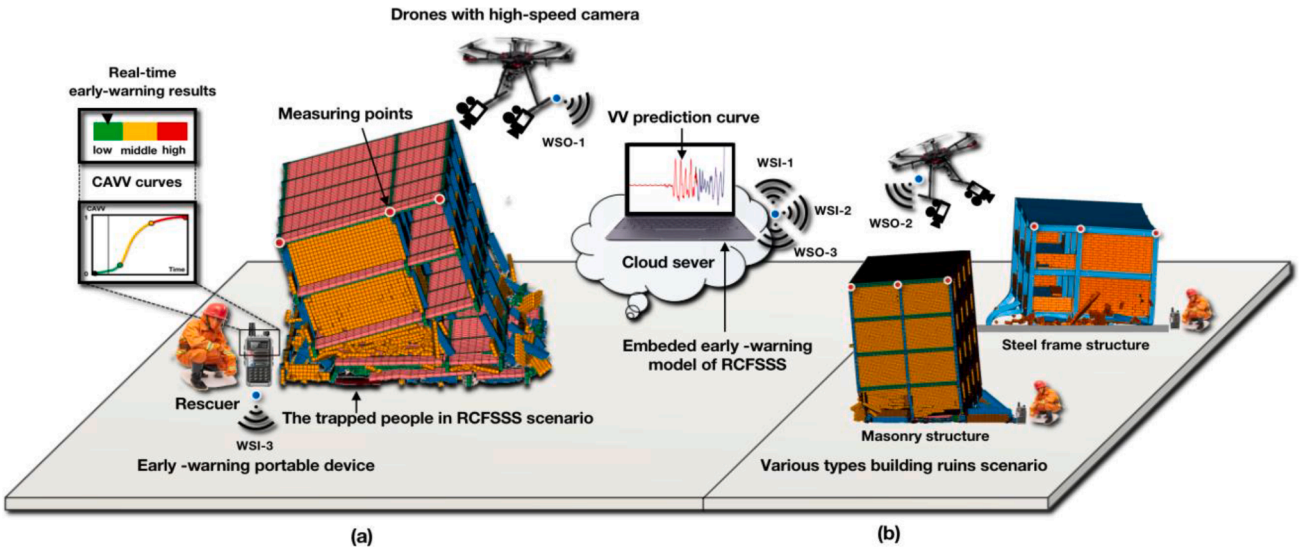


Fig. 13. Visualization of the early-warning application framework.

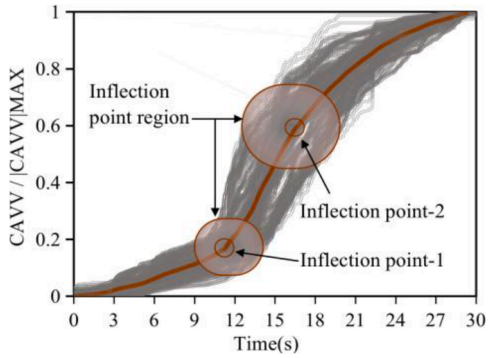


Fig. 14. The variety rule of the normalied CAVV curves.

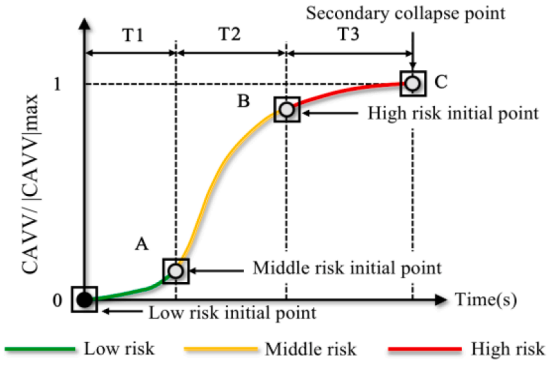


Fig. 15. Early-warning indicators of semi-ruined structures.

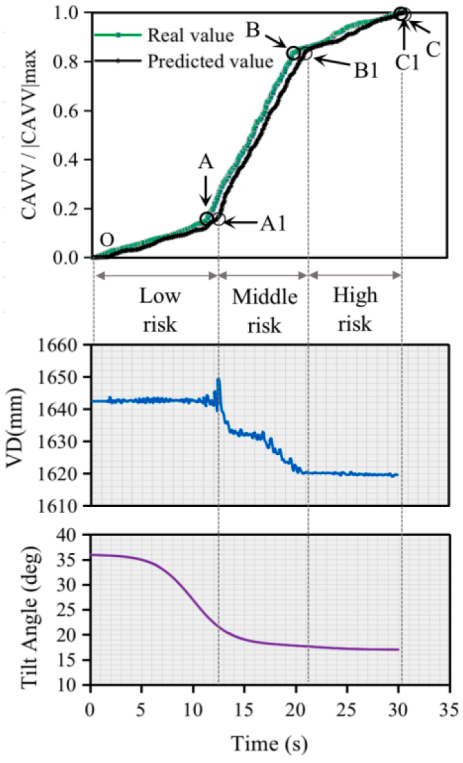


Fig. 16. CAVV curve of early-warning result.

masonry and steel frames. These findings highlight the role of deep learning, transfer learning, and domain-specific indicators in developing practical early-warning systems for complex structural scenarios.

5.4. Calculation cost of simulation

A large amount of simulation data was required for model training. The RCFSSS model was developed using a FEM-integrated FDEM approach, in which FEM simulated RC beams, slabs, and the ground, while FDEM modeled RC columns. To ensure accuracy in capturing the dynamic response of RCFSSS during aftershock-induced collapse, substantial computational resources were needed, making computational efficiency crucial to the method's feasibility.

Based on the 64 numerical models summarized in Table 1, the computational time for each model was analyzed (see Supplementary Table A. 5), and its relationship with the number of elements is shown in Fig. 17. The computational performance of the proposed FEM-integrated FDEM was also compared with that of the conventional FEM and AEM reported in previous studies (Botez, 2014; Grunwald, 2018; Khalil, 2020). For a similar number of elements, the proposed method shows markedly higher efficiency than the traditional FEM (e.g., models by Botez, Grunwald, and Khalil) but slightly lower efficiency than AEM (e.

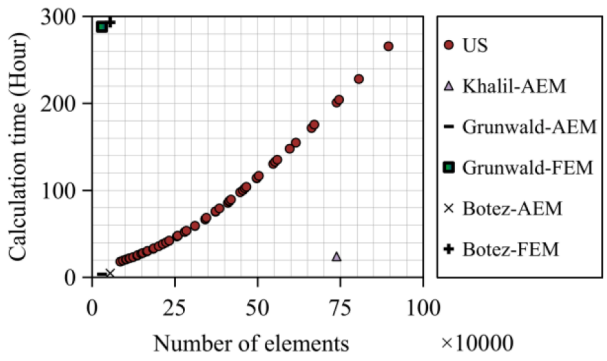


Fig. 17. The ralationship of element qualities and calculation time.

g., models by Botez and Grunwald). Moreover, the FEM-integrated FDEM implemented on the LS-DYNA platform supports restart functionality, allowing simulations to be resumed and extended—such as incorporating additional rescue supports within collapsed debris—providing a solid foundation for future research. In contrast, the AEM implemented in the Extreme Loading for Structures (ELS) software lacks this capability.

5.5. Cross-structural transfer test

A leave-one-structure-out transfer test was conducted to evaluate the cross-structural adaptability of the proposed transfer-learning–guided DCBT model. Specifically, the bottom-frame masonry structure (BFMS-SR), which collapsed during the 2008 Wenchuan earthquake, was selected as the target structure (Fig. 18(a)). The validated BFMS-SR model (Xu, 2024) was adopted, and 100 aftershock records (see Supplementary Figure A.3) were applied to induce secondary collapse, as illustrated in Fig. 18(b).

The resulting dynamic response dataset (100 aftershocks × 9 measurement points × 3 response types = 2700 samples) was used as the target domain. The model trained on the RCFSSS source domain was fine-tuned for the BFMS-SR target domain using Transfer Strategy I, to examine its ability to generalize learned seismic response features across structures. The predictive performance of the fine-tuned model on VV time-history curves was evaluated, and the results are summarized in Table 8.

These results confirm that the proposed transfer-learning–guided DCBT model maintains high predictive accuracy and generalization capability even when transferred across different structural types, demonstrating its robustness and potential applicability to masonry and other non-RC structures.

5.6. Limitations and outlook

In addition to structural dynamic response data, image-based datasets from UAVs or rescue robots are crucial for capturing concrete cracking, rebar exposure, and overall RCFSSS deformation under aftershocks (Song et al., 2024). Although the DCBT model demonstrates strong predictive capability, practical deployment faces challenges including computational efficiency, sensor noise, real-time data transmission, and user usability (Houghton et al., 2024). To date, validation has been limited to controlled laboratory conditions, highlighting the need for testing in realistic rescue-training environments.

Future research should enhance dataset diversity through large-scale field monitoring and multi-modal data fusion, and develop lightweight, adaptive deep-learning models suitable for wearable deployment. Comprehensive validation in real ruins and training scenarios will be essential to establish the reliability and operational value of the proposed early-warning framework.

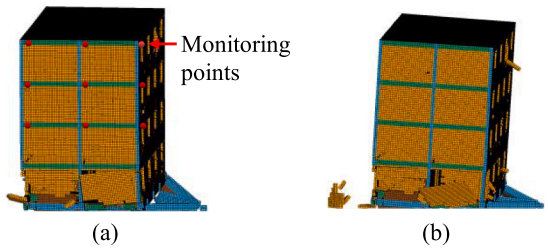


Fig. 18. The secondary collapse process of BFMS-SR model: (a) BFMS-SR and (b) the secondary collapse of BFMS-SR.

Table 8
The evaluation indices of BFMS-SR model.

MSE	*RMSE*	*MAE*	R^2	*PCC*	*MAPE*	*SMAPE*	*RMSPE*
9839	113.4	35.9	0.88	0.92	31.75	26.78	37.35

6. Conclusion

This study develops a transfer-learning–guided DCBT model for early warning of secondary collapse in semi-ruined RC frame structures (RCFSSS) under aftershocks. The proposed model achieves high predictive accuracy, with $RMSE \approx 112$ mm, $MAE \approx 36$ mm, $R^2 = 0.875$, and $PCC = 0.93$, outperforming all baseline and hybrid models. Transfer-learning strategies further enhance stability, maintaining $R^2 > 0.94$ under 5–10 % perturbations and reducing prediction errors by 15–30 % compared with training from scratch. Converting predicted VV into CAVV curves enables interpretable risk thresholds, where two key inflection points (≈1160 mm/s and ≈5035 mm/s) correspond to medium- and high-risk transitions. Overall, the proposed framework enables reliable, generalizable, and field-deployable early-warning assessments for post-earthquake rescue operations.

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Conflicts of interest

All authors approve this submission and declare no conflicts of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cacaie.2026.100001.

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